MILITARY SPECIFICATION

FASTENER, QUICK RELEASE, ALUMINUM

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cast aluminum, quick release fastener.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-591 - Aluminum Alloy Die Castings
QQ-A-596 - Aluminum Alloy Permanent and Semipermanent Mold Castings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5340

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QQ-A-601 - Aluminum-Alloy Sand Castings
QQ-S-698 - Steel, Sheet and Strip, Low Carbon
QQ-S-763 - Steel Bars, Wires, Shapes, and Forgings, Corrosion Resisting
QQ-W-321 - Wire, Copper Alloy
QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated)
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK, RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

4-1-98 - Fastener; Quick Release, Aluminum

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials.

3.2.1 Aluminum casting. Aluminum castings shall conform to any one of the following methods.

a. Sand castings shall conform to alloy No. 214 F or 356-T6 of QQ-A-601.

b. Die castings shall conform to alloy No. 360 of QQ-A-591.

C. Permanent mold castings shall conform to alloy No. 356 of QQ-A-596.

3.2.2 Steel wire. The steel wire shall conform to steel numbers 1010, 1015, or 1020, finish 1 of QQ-W-461.

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3.2.3 Steel, sheet and strip. The steel sheet shall conform to hot or cold rolled commercial quality of QQ-S-698. The steel strip shall conform to hot rolled commercial quality or cold rolled quarter hard no. 3 temper of QQ-S-698.

3.2.4 Corrosion-resisting steel. The corrosion-resisting steel shall conform to class 302 of QQ-S-763.

3.2.5 Copper alloy wire. The copper alloy wire shall conform to alloy No. 510 of QQ-W-321.

3.3 Construction. The construction of the fastener shall conform to the design, shape, and dimensions shown on Drawing 4-1-98 and as specified herein.

3.3.1 Body and latch assembly. The body and latch shall be cast from aluminum as specified in 3.2.1.

3.3.2 Ball bearings. The ball bearings shall be fabricated from corrosion-resisting steel as specified in 3.2.4.

3.3.3 Springs. The springs shall be fabricated from copper alloy wire as specified in 3.2.5.

3.3.4 Pin. The pin shall be fabricated from steel wire as specified in 3.2.2.

3.3.5 Plugs. The plugs shall be fabricated from steel sheet or strip as specified in 3.2.3 or steel wire as specified in 3.2.2.

3.3.6 Link. The link shall be cast from aluminum as specified in 3.2.1.

* 3.4 Finish. The steel pin and plugs shall be given a coat of phosphate treated zinc plate conforming to type IV, class Fe/Zn 13 Sc 3 of ASTM B 633.

3.5 Marking for identification. The fasteners shall be marked for identification in accordance with MIL-STD-130. The identification marks shall be incorporated in such a manner that the surfaces opposite the markings shall show no signs of penetration or unevenness or cause distortion in any part of the hardware.

3.6 Workmanship. The finished fasteners shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual. nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of complete quick release fasteners. The sample unit shall be one fastener. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 10.0 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish (steel pins and plug)	Not zinc plated		201
Construction and workmanship	Design or type not as specified	101	
	Blow holes, porosity, hard spots shrinkage, or cracks	102	
	Sharp edges not removed	103	
	Less than five coils in spring	104	
Assembly	Any component missing or incorrectly assembled	105	
	Latch cannot be snaplocked in body with both retaining balls showing tension	106	
Identification marking	Missing. incomplete, illegible, or incorrect		202

* 4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions annotated with an asterisk on Drawing 4-1-98. Any annotated dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of fasteners. The sample unit shall be one fastener. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit pack. The link shall be removed from the fastener and the latch shall be reset in the locked position. The link shall be positioned flat on the latch of the fastener with the larger end of the link at the hook

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end of the fastener body. Each fastener shall then be unit packed in a close-fitting clear polyethylene film bag of minimum 0.002-inch thickness. Closure shall be effected by heat sealing. Prior to or during the heat sealing closure operation, excess air within the bag shall be expelled.

5.1.1.2 Intermediate pack. Fifty quick release fasteners, unit packed as specified in 5.1.1.1, shall be packed in a snug-fitting intermediate fiberboard box conforming to style RSC, type CF or SF, class domestic, variety and grade optional of PPP-B-636. Closure shall be in accordance with the appendix of PPP-B-636.

* 5.1.2 Commercial. Quick release fasteners shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Quick release fasteners, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, water proofing, and reinforcing with flat steel strapping or tape banding shall be in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Quick release fasteners, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic. variety optional, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of PPP-B-636.

* 5.2.2.1 Weather resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Quick release fasteners, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), quick release fasteners, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

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* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The quick release fastener is used as a component of the single trace type man's sled harness.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant fiberboard boxes are required for level B shipment (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Subject term (key word) listing.

Equipment, accessory
Fastener
Harness
Quick release

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-F-11698D(GL)	Fastener, Quick Release, Aluminum

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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